

Science

Nature Notebook
Nature Walks & Scouting
Astronomy Lessons
Astronomy Labs





About the Course

This course includes the following topic(s): Astronomy Lessons, Astronomy Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Nature Notebook: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Astronomy Lessons

An elective in Earth Science, Astronomy is a fascinating topic in which learners explore celestial objects, history, and philosophy. Teachers wishing for explicit instruction on topics related to world view should include the optional Crossroads text.

About Astronomy Labs

Labs/fieldwork are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.



Placement & Combining Tips

Nature Notebook: Grades 9-12

Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

Astronomy Lessons

Recommended for students in Grades 11-12, depending on math placement. Students should have completed Algebra 1 (Algebra 2 if taking Denison Success) and Geometry.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Nature Notebook: Grades 9-12 1+ time/week 20 min+	
9-12	Nature Walks & Scouting: Grades 9-12 1 time/week 30 min+	
11-12	Astronomy Lessons 5 times/week 45 min	100 Things to See in the Night Sky The Last Stargazers Astronomy Textbook The Day Without Yesterday The Crossroads of Science and Faith: Answers to Exercises The Crossroads of Science and Faith: Astronomy Through a Christian Worldview
11-12	Astronomy Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Astronomy				
Astronomy Lessons Nature Walks & Scouting: Grades 9-12	Astronomy Lessons	Astronomy Lessons	Astronomy Lessons Nature Notebook: Grades 9-12	Astronomy Lessons Astronomy Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Astronomy Lessons

- ☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.
- ☐ Obtain materials from the supply lists.
- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Science: Astronomy

Astronomy Lessons



100 Things to See in the Night Sky



The Last Stargazers



Astronomy Textbook



The Day Without Yesterday



The Crossroads of Science and Faith: Answers to Exercises



The Crossroads of Science and Faith: Astronomy Through a Christian Worldview



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

Science: Astronomy

[Click THIS text or scan the QR code for links.](#)

QR

Astronomy Lessons

- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [Stellarium](#)
- ∞ [Hubble Space Telescope](#)
- ∞ [Webb Space Telescope](#)
- ∞ [Earth Sky](#)
- ∞ [Astronomy Picture of the Day](#)
- ∞ [Veritas Forum](#)
- ∞ [Galaxy Zoo](#)
- ∞ [Globe at Night](#)
- ∞ [Exoplanet Watch](#)
- ∞ [Space Math](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [Crossroads Answer Key Part 1](#)
- ∞ [Crossroads Answer Key Part 2](#)

Science: Astronomy

How To Approach



Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

Science: Astronomy

How To Approach



Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Term 1

WEEK 1

45m Astronomy Lessons - Lesson 1

Identification

Materials: Astronomy Text, star finder

→ INTRO

This is your day each week to prepare for your field studies. You will begin with a short reading, but most of the lesson will be used for planning. To find a local club, explore your locale at the links below:

- ∞ Link: Night Sky Network
- ∞ Link: New Planetarium
- ∞ Link: Royal Astronomical Society

→ READ, NARRATE, & DISCUSS

Unit I Lesson 1 through the section "Shifting Skies" (to "in the sky.")

→ PREP

- Learn to orient yourself with the star finder.
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Note what time the sky becomes dark enough to see the stars.
- Check the weather and note which nights will have cloud cover so that you can plan a night each week to do some sky gazing.
- As you are able, seek out local observatories or clubs to increase your knowledge and experience locating objects in the sky.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 1

45m Astronomy Lessons - Lesson 2

How Astronomy Works, Highlights from History

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 2. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 1

45m Astronomy Lessons - Lesson 3

Astronomy Literature

Materials: The Last Stargazers

→ INTRO

Two days each week will be spent reading important works of astronomical literature. The lesson plans give you an idea of how many pages to read each week to finish the book on schedule, but this is flexible. If a book takes longer than suggested, you might read outside the lesson time or even leave off without finishing it. If you read faster than suggested, use the extra time for other activities in the course.

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 1

45m Astronomy Lessons - Lesson 4

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 1

45m Astronomy Lessons - Lesson 5

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ INTRO

About half of these days will be spent reading current events; the other half will be spent participating in citizen science. Keep a log or journal entry of your findings and thoughts in your Astronomy Notebook.

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here.

OR

Term 1

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links).

WEEK 1

60m Astronomy Labs - Lesson 1

Materials: journal, star finder

→ OBSERVE

Simply notice on this first night of observing. What are your first impressions? Or are you able to relocate any objects previously found? Do you notice anything new based on your use of the star finder or Quick Links? Keep a record of what you observe.

WEEK 2

45m Astronomy Lessons - Lesson 6

Identification

Materials: 100 Things to See in the Night Sky

→ READ, NARRATE, & DISCUSS

How to Use This Book

→ PREP

- Learn to orient yourself by locating or reminding yourself where the four cardinal directions point in your place.
- Practice measuring objects at a distance using your hands.
- Review your viewing experience last week. Was it a good spot? Are there other possibilities that would be better? Can you do anything to improve your experience?
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Check the weather and note which nights will have cloud cover so that you can plan a night each week to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 2

45m Astronomy Lessons - Lesson 7

45m Astronomy - Lesson 7

Materials: Astronomy Text

How Astronomy Works, Gravity

→ READ, NARRATE, & DISCUSS

Unit 1 Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 2

45m Astronomy Lessons - Lesson 8

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 2

45m Astronomy Lessons - Lesson 9

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 2

45m Astronomy Lessons - Lesson 10

45m Astronomy - Lesson 10

Materials: computer, optional: Crossroads of Science and Faith text

Astronomy Current Events and Research

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

Term 1

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 2

60m Astronomy Labs - Lesson 2

Materials: journal, star finder

→ OBSERVE

Notice any objects that are or are becoming familiar. In what directions do they currently lie? Using your hands, measure their locations in the sky. Do these locations change during the evening? Keep a record of what you observe.

★ STUDENT & TEACHER TIP
Note that next week's fieldwork should be done in one or more daytime sessions!

WEEK 3

45m Astronomy Lessons - Lesson 11

Identification

Materials: 100 Things to See in the Night Sky, binoculars, paper, tripod/stand for binoculars

→ READ, NARRATE, & DISCUSS

Part I: Introduction & The Sun

→ PREP

- Review where the four cardinal directions point in your place.
- Practice measuring objects at a distance using your hands.
- You will view during the day this week - check the weather and decide when you will do some sky gazing. Will you view it at sunrise? Daytime? Sunset? Since most of the work in this course will be at night, make the most of this week!
- Ensure that you understand how to set up your binoculars securely, so that you can project the sun for safe viewing:
 - ∞ Website Link: Track Activity on the Surface of the Sun
 - ∞ Video Link: How to Project the Sun with Binoculars
- If you would like to consider an equinox special study, check out the idea below:
 - ∞ Website Link: Sun Shadows

• PLAN WEEKLY
nature walk - record observations
science free read

WEEK 3

45m Astronomy Lessons - Lesson 12

How Astronomy Works, Spatial Relations

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 4. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 3

45m Astronomy Lessons - Lesson 13

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 3

45m Astronomy Lessons - Lesson 14

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 3

45m Astronomy Lessons - Lesson 15

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

Term 1

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 3

60m Astronomy Labs - Lesson 3

Materials: journal, binoculars, paper, tripod/stand for binoculars

→ OBSERVE

Use your hands to measure locations and your binocular set-up to observe the sun's projection. DO NOT LOOK DIRECTLY AT THE SUN! Keep a record of what you observe.

WEEK 4

45m Astronomy Lessons - Lesson 16

Identification

Materials: 100 Things to See in the Night Sky

→ READ, NARRATE, & DISCUSS

The Moon

→ PREP

- Practice measuring objects at a distance using your hands.
- Review your previous nighttime viewing experience. Can you do anything to improve your experience?
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Check the weather and note which nights will have cloud cover so that you can plan a night week to do some sky gazing.
- Note when the moon will rise and set this week to plan for viewing.
- If you would like to consider a lunar special study, you might journal the phases over the next month. For a more challenging study, check out the idea below:
∞ Link: Isabel Williamson Lunar Observing Program

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 4

45m Astronomy Lessons - Lesson 17

How Astronomy Works, Modern Astronomy

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 5. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 4

45m Astronomy Lessons - Lesson 18

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 4

45m Astronomy Lessons - Lesson 19

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 4

45m Astronomy Lessons - Lesson 20

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

Term 1

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 4

60m Astronomy Labs - Lesson 4

Materials: journal, star finder, binoculars/telescope

→ OBSERVE

Use your hands to measure locations of the moon and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope to view the moon up close. Keep a record of what you observe.

WEEK 5

45m Astronomy Lessons - Lesson 21

45m Astronomy - Lesson 21

Materials: 100 Things to See in the Night Sky

Identification

→ READ, NARRATE, & DISCUSS

The Naked-Eye Planets & Mercury

→ PREP

- Practice measuring objects at a distance using your hands.
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings. For example, if Jupiter is visible next week, then consider reading about Jupiter next week and come back to the others later.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5

45m Astronomy Lessons - Lesson 22

How Astronomy Works, Electromagnetic Waves

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 6. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 5

45m Astronomy Lessons - Lesson 23

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 5

45m Astronomy Lessons - Lesson 24

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 5

45m Astronomy Lessons - Lesson 25

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

Term 1

WEEK 5	60m Astronomy Labs - Lesson 5 Materials: journal, star finder, binoculars/telescope → OBSERVE Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.	
WEEK 6	45m Astronomy Lessons - Lesson 26 <i>Identification</i> Materials: 100 Things to See in the Night Sky → READ, NARRATE, & DISCUSS Venus → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 6	45m Astronomy Lessons - Lesson 27 <i>How Astronomy Works, Frequency and Wavelength</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit 1 Lsn 7. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.	
WEEK 6	45m Astronomy Lessons - Lesson 28 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 6	45m Astronomy Lessons - Lesson 29 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 6	45m Astronomy Lessons - Lesson 30 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 6	60m Astronomy Labs - Lesson 6 Materials: journal, star finder, binoculars/telescope → OBSERVE Use your hands to measure locations of the moon, planets, and other objects noted with your star	

Term 1

finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 7

45m Astronomy Lessons - Lesson 31

Identification

Materials: 100 Things to See in the Night Sky

→ READ, NARRATE, & DISCUSS

Mars

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7

45m Astronomy Lessons - Lesson 32

How Astronomy Works, Spectral Analysis

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 8. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 7

45m Astronomy Lessons - Lesson 33

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 7

45m Astronomy Lessons - Lesson 34

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 7

45m Astronomy Lessons - Lesson 35

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 7

60m Astronomy Labs - Lesson 7

Materials: journal, star finder, binoculars/telescope

→ OBSERVE

Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

Term 1

WEEK 8	45m Astronomy Lessons - Lesson 36 <i>Identification</i> Materials: 100 Things to See in the Night Sky → READ, NARRATE, & DISCUSS Jupiter → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 8	45m Astronomy Lessons - Lesson 37 <i>How Astronomy Works, The Grand and Magnificent Universe</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit 1 Lsn 9. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.	
WEEK 8	45m Astronomy Lessons - Lesson 38 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 8	45m Astronomy Lessons - Lesson 39 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 8	45m Astronomy Lessons - Lesson 40 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 8	60m Astronomy Labs - Lesson 8 Materials: journal, star finder, binoculars/telescope → OBSERVE Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.	
WEEK 9	45m Astronomy Lessons - Lesson 41 <i>Identification</i> Materials: 100 Things to See in the Night Sky → READ, NARRATE, & DISCUSS Saturn	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

Term 1

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 9

45m Astronomy Lessons - Lesson 42

How Astronomy Works, The Grand Design

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 10. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 9

45m Astronomy Lessons - Lesson 43

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 9

45m Astronomy Lessons - Lesson 44

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 9

45m Astronomy Lessons - Lesson 45

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 9

60m Astronomy Labs - Lesson 9

Materials: journal, star finder, binoculars/telescope

→ OBSERVE

Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 10

45m Astronomy Lessons - Lesson 46

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Part II: Introduction, The Northern Sky, & Ursa Major

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note where Ursa Major will be.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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WEEK 10 45m Astronomy Lessons - Lesson 47

How Astronomy Works, The Telescope

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 11. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 10 45m Astronomy Lessons - Lesson 48

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 10 45m Astronomy Lessons - Lesson 49

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 10 45m Astronomy Lessons - Lesson 50

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 10 60m Astronomy Labs - Lesson 10

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of Ursa Major, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 11 45m Astronomy Lessons - Lesson 51

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Merak and Dubhe & Mizar and Alcor

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of Ursa Major and the associated stars.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11 45m Astronomy Lessons - Lesson 52

How Astronomy Works, Space Exploration

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit I Lsn 12. Narrate silently to yourself or write. Copy any diagrams and make sketches if

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helpful to understand the text. Consider any questions from the text as time permits.

WEEK 11 **45m Astronomy Lessons - Lesson 53** *Astronomy Catch-up Day*

→ CATCH-UP

Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11 **45m Astronomy Lessons - Lesson 54** *Astronomy Catch-up Day*

→ CATCH-UP

Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11 **45m Astronomy Lessons - Lesson 55** *Astronomy Current Events and Research*

 Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 11 **60m Astronomy Labs - Lesson 11** Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of Ursa Major and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 12 **45m Astronomy Lessons - Lesson 56** *Exam Week*

→ EXAMS

Answer question(s) related to the course.

WEEK 12 **45m Astronomy Lessons - Lesson 57** *Exam Week*

→ EXAMS

Answer question(s) related to the course.

WEEK 12 **45m Astronomy Lessons - Lesson 58** *Exam Week*

→ EXAMS

Answer question(s) related to the course.

WEEK 12 **45m Astronomy Lessons - Lesson 59** *Exam Week*

→ EXAMS

Answer question(s) related to the course.

WEEK 12 **45m Astronomy Lessons - Lesson 60** *Exam Week*

→ EXAMS

Answer question(s) related to the course.

WEEK 12 **60m Astronomy Labs - Lesson 12** *Exam Week*

→ EXAMS

Answer question(s) related to the course.

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WEEK 1	45m Astronomy Lessons - Lesson 61 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Ursa Minor, Polaris, & Kochab and Pherkad → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of Ursa Minor and the associated stars. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 1	45m Astronomy Lessons - Lesson 62 <i>How Astronomy Works, The Solar System</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit II Lsn 1. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.	
WEEK 1	45m Astronomy Lessons - Lesson 63 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 1	45m Astronomy Lessons - Lesson 64 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 1	45m Astronomy Lessons - Lesson 65 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → INTRO Use this time to explore current events in astronomy, keeping a weekly log or journal entry of your findings and thoughts in your Astronomy Notebook. → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links).	
WEEK 1	60m Astronomy Labs - Lesson 13 Materials: journal, star finder → OBSERVE Use your hands to measure locations of Ursa Major, Ursa Minor, and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Has the location of Ursa Major changed from your last viewing? Do the locations change during your session? Keep a record of what you observe.	

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WEEK 2	45m Astronomy Lessons - Lesson 66 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Cassiopeia & Caph → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of Cassiopeia and its associated star(s). • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 2	45m Astronomy Lessons - Lesson 67 <i>How Astronomy Works, Formation of Our Solar System</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit II Lsn 2. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.	
WEEK 2	45m Astronomy Lessons - Lesson 68 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 2	45m Astronomy Lessons - Lesson 69 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 2	45m Astronomy Lessons - Lesson 70 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 2	60m Astronomy Labs - Lesson 14 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 3	45m Astronomy Lessons - Lesson 71 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Cepheus & Delta Cephei	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of Cepheus and its associated star(s).
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 3

45m Astronomy Lessons - Lesson 72

How Astronomy Works, The Sun

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit II Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore the Sun at NASA:

∞ Link: NASA - Sun

WEEK 3

45m Astronomy Lessons - Lesson 73

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 3

45m Astronomy Lessons - Lesson 74

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 3

45m Astronomy Lessons - Lesson 75

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 3

60m Astronomy Labs - Lesson 15

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 4

45m Astronomy Lessons - Lesson 76

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Draco & Thuban

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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- Use the star finder to note the position of Draco and its associated star(s).
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 4

45m Astronomy Lessons - Lesson 77

How Astronomy Works, Earth & Moon

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit II Lsn 4-5. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore Earth and the Moon at NASA:

∞ Link: NASA - Solar System

WEEK 4

45m Astronomy Lessons - Lesson 78

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 4

45m Astronomy Lessons - Lesson 79

Astronomy Literature

Materials: The Last Stargazers

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 4

45m Astronomy Lessons - Lesson 80

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 4

60m Astronomy Labs - Lesson 16

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 5

45m Astronomy Lessons - Lesson 81

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Now move to whatever season you are in and continue reading.

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of the stars you are interested in.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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WEEK 5  45m Astronomy Lessons - Lesson 82 <i>How Astronomy Works, Comets</i>  Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit II Lsn 6. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW Explore comets at NASA: ∞ Link: NASA - Solar System	
WEEK 5  45m Astronomy Lessons - Lesson 83 <i>Astronomy Literature</i>  Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 5  45m Astronomy Lessons - Lesson 84 <i>Astronomy Literature</i>  Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 5  45m Astronomy Lessons - Lesson 85 <i>Astronomy Current Events and Research</i>  Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 5  60m Astronomy Labs - Lesson 17  Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 6  45m Astronomy Lessons - Lesson 86 <i>Identification</i>  Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of the stars you are interested in. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY  nature walk - record observations  science free read

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WEEK 6	45m Astronomy Lessons - Lesson 87 <i>How Astronomy Works, Asteroids and Meteors</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit II Lsn 7 - half of Lsn 8. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW Explore asteroids at NASA: ∞ Link: NASA - Solar System	
WEEK 6	45m Astronomy Lessons - Lesson 88 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 6	45m Astronomy Lessons - Lesson 89 <i>Astronomy Literature</i> Materials: The Last Stargazers → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 6	45m Astronomy Lessons - Lesson 90 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 6	60m Astronomy Labs - Lesson 18 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 7	45m Astronomy Lessons - Lesson 91 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of the stars you are interested in. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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WEEK 7

45m Astronomy Lessons - Lesson 92

How Astronomy Works, Meteors and Impactors

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit II, the remaining half of Lsn 8 - Lsn 9. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore meteors at NASA:
∞ Link: NASA - Solar System

WEEK 7

45m Astronomy Lessons - Lesson 93

Astronomy Literature

Materials: The Last Stargazers, optional: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday.

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 7

45m Astronomy Lessons - Lesson 94

Astronomy Literature

Materials: The Last Stargazers, optional: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday.

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 7

45m Astronomy Lessons - Lesson 95

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 7

60m Astronomy Labs - Lesson 19

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 8

45m Astronomy Lessons - Lesson 96

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of the stars you are interested in.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

- **PLAN WEEKLY**
 - ☐ nature walk - record observations
 - ☐ science free read

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WEEK 8	45m Astronomy Lessons - Lesson 97 <i>How Astronomy Works, Highlights of the Planets</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit II Lsn 10. If you want to break this chapter up, feel free to stop when full; you may pick up next week. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA: ∞ Link: NASA - Solar System	
WEEK 8	45m Astronomy Lessons - Lesson 98 <i>Astronomy Literature</i> Materials: The Last Stargazers, optional: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday. → WRITE Write a narration in your Astronomy Notebook.	
WEEK 8	45m Astronomy Lessons - Lesson 99 <i>Astronomy Literature</i> Materials: The Last Stargazers, optional: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday. → WRITE Write a narration in your Astronomy Notebook.	
WEEK 8	45m Astronomy Lessons - Lesson 100 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 8	60m Astronomy Labs - Lesson 20 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 9	45m Astronomy Lessons - Lesson 101 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of the stars you are interested in. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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WEEK 9	45m Astronomy Lessons - Lesson 102 <i>How Astronomy Works, The Planets Continued</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Continue Unit II Lsn 10. as needed. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA: ∞ Link: NASA - Solar System	
WEEK 9	45m Astronomy Lessons - Lesson 103 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). There will be some math and science in the story that you have not yet learned, but that is okay. The point is really about the story. → WRITE Write a narration in your Astronomy Notebook.	
WEEK 9	45m Astronomy Lessons - Lesson 104 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 9	45m Astronomy Lessons - Lesson 105 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 9	60m Astronomy Labs - Lesson 21 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 10	45m Astronomy Lessons - Lesson 106 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of the stars you are interested in. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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WEEK 10	45m Astronomy Lessons - Lesson 107 <i>How Astronomy Works, The Planets Continued</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Continue Unit II Lsn 10. as needed. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA: ∞ Link: NASA - Solar System	
WEEK 10	45m Astronomy Lessons - Lesson 108 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 10	45m Astronomy Lessons - Lesson 109 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 10	45m Astronomy Lessons - Lesson 110 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 10	60m Astronomy Labs - Lesson 22 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 11	45m Astronomy Lessons - Lesson 111 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of the stars you are interested in. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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WEEK 11  45m Astronomy Lessons - Lesson 112 <i>How Astronomy Works, The Planets Continued</i>  Materials: Astronomy Text → READ, NARRATE, & DISCUSS Continue Unit II Lsn 10. as needed. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA: ∞ Link: NASA - Solar System	
WEEK 11  45m Astronomy Lessons - Lesson 113 <i>Astronomy Catch-up Day</i> → CATCH-UP Use this time to complete any readings, note-booking, or projects for the course.	
WEEK 11  45m Astronomy Lessons - Lesson 114 <i>Astronomy Catch-up Day</i> → CATCH-UP Use this time to complete any readings, note-booking, or projects for the course.	
WEEK 11  45m Astronomy Lessons - Lesson 115 <i>Astronomy Current Events and Research</i>  Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 11  60m Astronomy Labs - Lesson 23  Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 12  45m Astronomy Lessons - Lesson 116 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12  45m Astronomy Lessons - Lesson 117 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12  45m Astronomy Lessons - Lesson 118 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12  45m Astronomy Lessons - Lesson 119 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	



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WEEK 12 45m Astronomy Lessons - Lesson 120 Exam Week → EXAMS Answer question(s) related to the course.	
WEEK 12 60m Astronomy Labs - Lesson 24 Exam Week → EXAMS Answer question(s) related to the course.	

Term 3

WEEK 1

45m Astronomy Lessons - Lesson 121

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of the stars you are interested in.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 1

45m Astronomy Lessons - Lesson 122

How Astronomy Works, Cosmology

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit III Lsn 1. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 1

45m Astronomy Lessons - Lesson 123

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 1

45m Astronomy Lessons - Lesson 124

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 1

45m Astronomy Lessons - Lesson 125

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ

Use this time to explore current events in astronomy, keeping a weekly log or journal entry of your findings and thoughts in your Astronomy Notebook.

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links).

WEEK 1

60m Astronomy Labs - Lesson 25

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

Term 3

WEEK 2	45m Astronomy Lessons - Lesson 126 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP • Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. • Use the star finder to note the position of the stars you are interested in. • Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 2	45m Astronomy Lessons - Lesson 127 <i>How Astronomy Works, Cosmology, and Theology</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit III Lsn 2. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.	
WEEK 2	45m Astronomy Lessons - Lesson 128 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 2	45m Astronomy Lessons - Lesson 129 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 2	45m Astronomy Lessons - Lesson 130 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 2	60m Astronomy Labs - Lesson 26 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 3	45m Astronomy Lessons - Lesson 131 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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→ PREP

- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of the stars you are interested in.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 3

45m Astronomy Lessons - Lesson 132

How Astronomy Works, Beyond the Solar System

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit III half of Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

WEEK 3

45m Astronomy Lessons - Lesson 133

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 3

45m Astronomy Lessons - Lesson 134

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 3

45m Astronomy Lessons - Lesson 135

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 3

60m Astronomy Labs - Lesson 27

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 4

45m Astronomy Lessons - Lesson 136

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

Term 3

WEEK 4	45m Astronomy Lessons - Lesson 137 <i>How Astronomy Works, Beyond the Solar System</i> Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit III, the remaining half of Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.	
WEEK 4	45m Astronomy Lessons - Lesson 138 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 4	45m Astronomy Lessons - Lesson 139 <i>Astronomy Literature</i> Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 4	45m Astronomy Lessons - Lesson 140 <i>Astronomy Current Events and Research</i> Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 4	60m Astronomy Labs - Lesson 28 Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 5	45m Astronomy Lessons - Lesson 141 <i>Identification</i> Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP Use your Quick Links, star finder, and weather forecast to plan for sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read

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WEEK 5

45m Astronomy Lessons - Lesson 142

How Astronomy Works, The Life Cycle of a Star

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit III, half of Lsn 4. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

WEEK 5

45m Astronomy Lessons - Lesson 143

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 5

45m Astronomy Lessons - Lesson 144

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 5

45m Astronomy Lessons - Lesson 145

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 5

60m Astronomy Labs - Lesson 29

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 6

45m Astronomy Lessons - Lesson 146

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 6

45m Astronomy Lessons - Lesson 147

How Astronomy Works, The Life Cycle of a Star

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit III, the remaining half of Lsn 4. Narrate silently to yourself or write. Copy any diagrams and

Term 3

make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

WEEK 6

45m Astronomy Lessons - Lesson 148

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 6

45m Astronomy Lessons - Lesson 149

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 6

45m Astronomy Lessons - Lesson 150

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 6

60m Astronomy Labs - Lesson 30

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 7

45m Astronomy Lessons - Lesson 151

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7

45m Astronomy Lessons - Lesson 152

How Astronomy Works, Intrinsic Variables, and Determining Distance

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit III Lsn 5. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ VIEW

Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

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WEEK 7	45m Astronomy Lessons - Lesson 153 <i>Astronomy Literature</i>	Materials: The Day Without Yesterday	→ READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 7	45m Astronomy Lessons - Lesson 154 <i>Astronomy Literature</i>	Materials: The Day Without Yesterday	→ READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 7	45m Astronomy Lessons - Lesson 155 <i>Astronomy Current Events and Research</i>	Materials: computer, optional: Crossroads of Science and Faith text	→ READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 7	60m Astronomy Labs - Lesson 31	Materials: journal, star finder	→ OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 8	45m Astronomy Lessons - Lesson 156 <i>Identification</i>	Materials: 100 Things to See in the Night Sky, star finder	→ READ, NARRATE, & DISCUSS Continue reading. → PREP Use your Quick Links, star finder, and weather forecast to plan for sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 8	45m Astronomy Lessons - Lesson 157 <i>How Astronomy Works, Ways to Determine Distances</i>	Materials: Astronomy Text	→ READ, NARRATE, & DISCUSS Unit III Lsn 6. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → VIEW Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.	
WEEK 8	45m Astronomy Lessons - Lesson 158 <i>Astronomy Literature</i>	Materials: The Day Without Yesterday	→ READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days).	

Term 3

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 8

45m Astronomy Lessons - Lesson 159

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 8

45m Astronomy Lessons - Lesson 160

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 8

60m Astronomy Labs - Lesson 32

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 9

45m Astronomy Lessons - Lesson 161

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9

45m Astronomy Lessons - Lesson 162

How Astronomy Works, The H-R Diagram

Materials: Astronomy Text

→ READ, NARRATE, & DISCUSS

Unit III Lsn 7. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

→ PROJECT

Use any additional time to work on any projects (e.g., special studies, citizen science, etc.).

WEEK 9

45m Astronomy Lessons - Lesson 163

Astronomy Literature

Materials: The Day Without Yesterday

→ READ, NARRATE, & DISCUSS

Read about 18 pages per week (over 2 lesson days).

→ WRITE

Write a narration in your Astronomy Notebook.

Term 3

WEEK 9	45m Astronomy Lessons - Lesson 164 <i>Astronomy Literature</i>	Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 9	45m Astronomy Lessons - Lesson 165 <i>Astronomy Current Events and Research</i>	Materials: computer, optional: Crossroads of Science and Faith text → READ CURRENT EVENTS EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover. OR → PRACTICE CITIZEN SCIENCE Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.	
WEEK 9	60m Astronomy Labs - Lesson 33	Materials: journal, star finder → OBSERVE Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.	
WEEK 10	45m Astronomy Lessons - Lesson 166 <i>Identification</i>	Materials: 100 Things to See in the Night Sky, star finder → READ, NARRATE, & DISCUSS Continue reading. → PREP Use your Quick Links, star finder, and weather forecast to plan for sky gazing.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 10	45m Astronomy Lessons - Lesson 167 <i>How Astronomy Works, Other Planetary Systems</i>	Materials: Astronomy Text → READ, NARRATE, & DISCUSS Unit III Lsn 8. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits. → PROJECT Use any additional time to work on any projects (e.g., special studies, citizen science, etc.).	
WEEK 10	45m Astronomy Lessons - Lesson 168 <i>Astronomy Literature</i>	Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days). → WRITE Write a narration in your Astronomy Notebook.	
WEEK 10	45m Astronomy Lessons - Lesson 169 <i>Astronomy Literature</i>	Materials: The Day Without Yesterday → READ, NARRATE, & DISCUSS Read about 18 pages per week (over 2 lesson days).	

Term 3

→ WRITE

Write a narration in your Astronomy Notebook.

WEEK 10

45m Astronomy Lessons - Lesson 170

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 10

60m Astronomy Labs - Lesson 34

Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 11

45m Astronomy Lessons - Lesson 171

Identification

Materials: 100 Things to See in the Night Sky, star finder

→ READ, NARRATE, & DISCUSS

Continue reading.

→ PREP

Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11

45m Astronomy Lessons - Lesson 172

Astronomy Catch-up Day

→ CATCH-UP

Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11

45m Astronomy Lessons - Lesson 173

Astronomy Catch-up Day

→ CATCH-UP

Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11

45m Astronomy Lessons - Lesson 174

Astronomy Catch-up Day

→ CATCH-UP

Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11

45m Astronomy Lessons - Lesson 175

Astronomy Current Events and Research

Materials: computer, optional: Crossroads of Science and Faith text

→ READ CURRENT EVENTS

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

→ PRACTICE CITIZEN SCIENCE

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

Term 3

WEEK 11 ☐ 60m Astronomy Labs - Lesson 35

☐ Materials: journal, star finder

→ OBSERVE

Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 12 ☐ 45m Astronomy Lessons - Lesson 176

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 ☐ 45m Astronomy Lessons - Lesson 177

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 ☐ 45m Astronomy Lessons - Lesson 178

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 ☐ 45m Astronomy Lessons - Lesson 179

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 ☐ 45m Astronomy Lessons - Lesson 180

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 ☐ 60m Astronomy Labs - Lesson 36

Exam Week

→ EXAMS

Answer question(s) related to the course.

Science: Astronomy

Examination

Term 1

Astronomy Lessons

- 100 Things to See: Tell something you learned about observing the sun, moon, or planets. OR Tell about stargazing this term. For either question, share your notebook. What did you observe? What questions did you think about?
- Astronomy: Explain how we can know that something exists if we cannot see it (such as wind and black holes)?
- Astronomy: Why is light important to astronomy? What is light? What is the Doppler effect?
- The Last Stargazers: Tell about something that surprised you about the life and work of an astronomer from what you have read so far. OR, Many people see science and art as very different pursuits, but the author tells about scientists who love music, poetry, drawing, etc. Explain how you think science and art work together as a person explores the Things around them.
- Tell about a current event or citizen science project from this term.

Term 2

Astronomy Lessons

- 100 Things to See: Explain how to find some of the major constellations, such as Ursa Major/minor, Cassiopeia, Draco, etc. OR Tell about stargazing this term. For either question, share your notebook. What did you observe? What questions did you think about?
- Astronomy: Tell about 3 unique or interesting features found among objects in the solar system.
- The Last Stargazers: Tell about a scientific aspect of astronomy (the discoveries, the mysteries, the science of telescopes, etc) that you found interesting.
- The Day Without Yesterday: The scientific community persistently resisted the idea of an expanding universe. Give reasons for this from what you read.
- Tell about a current event or citizen science project from this term.

Term 3

Astronomy Lessons

- 100 Things to See: Tell about some of the seasonal stars you learned to find this term. OR Tell about stargazing this term. For either question, share your notebook. What did you observe? What questions did you think about?
- Astronomy: Explain what a galaxy, black hole, nebula, supernova, and binary star system is? How are they connected? OR Explain 4 ideas as to why the universe seems designed for life.
- Astronomy: Describe the life cycle of a star. OR What are some questions that theology and cosmology share?
- The Day Without Yesterday: Lemaître's work has been described as 'apologetics in action.' Explain.
- Tell about a current event or citizen science project from this term.